HDOC-49

Pivot Integer Using Pointers in C

Problem Statement

Write a C program that reads a positive integer n. Pass n to a function named **pivot_integer()** using its address. Use pointers to pass the value between the calling function and pivot_integer(). The function must find the pivot integer x between 1 and n such that the sum of integers from 1 to x, inclusive, equals the sum of integers from x to n, inclusive. Display x if it exists; otherwise display -1.

Concept Used

- Functions
- Pointers / call by reference
- Loops and conditional statements

Algorithm

1. Start.
2. Read a positive integer n in main().
3. Pass the address of n to pivot_integer() using a pointer.
4. For each possible x from 1 to n, calculate the sum from 1 to x and the sum from x to n.
5. If both sums are equal, return x as the pivot integer.
6. If no such x is found, return -1.
7. Display the returned value and stop.

C Program

```
#include <stdio.h>

int pivot_integer(int *n)
{
   int x, i;
   int leftSum, rightSum;

   for (x = 1; x <= *n; x++)
   {
      leftSum = 0;
      rightSum = 0;

      for (i = 1; i <= x; i++)
         leftSum += i;

      for (i = x; i <= *n; i++)
         rightSum += i;

      if (leftSum == rightSum)
         return x;
   }
```

```
    return -1;
}

int main(void)
{
    int n, result;

    printf("Enter a positive integer n: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    result = pivot_integer(&n);
    printf("Pivot integer: %d\n", result);

    return 0;
}
```

Test Cases

Test Case	Input n	Expected Output	Reason
1	8	Pivot integer: 6	1+2+3+4+5+6 = 6+7+8 = 21
2	4	Pivot integer: -1	No integer satisfies the condition
3	1	Pivot integer: 1	Both sums are 1
4	49	Pivot integer: 35	Sum 1 to 35 equals sum 35 to 49

Sample Output

```
Enter a positive integer n: 8
Pivot integer: 6
```

```
Enter a positive integer n: 4
Pivot integer: -1
```

Result

The program successfully passes n by address to pivot_integer(), uses a pointer to access its value, and displays the pivot integer when one exists; otherwise it displays -1.